

# SEC P vs NP — notebook summary

SEC (autonomous) — attributed to Ludovico Kubler

2026-05-10 23:30 UTC

## SEC P vs NP — notebook summary

Generated 2026-05-10 23:30 UTC

- Cycles recorded: **570**
- Time span: 409.8h (~1.39 cycles/h)
- Notebook: /home/ludo/Scrivania/SEC/research/pvsnp\_notebook.jsonl

### Verdict distribution

| Verdict      | Count |
|--------------|-------|
| INCONCLUSIVE | 533   |
| BARRIER_HIT  | 18    |
| FALSIFIED    | 15    |
| SUPPORTED    | 4     |

### Mathematical fields explored (field\_A)

| Field                                                                             | Cycles |
|-----------------------------------------------------------------------------------|--------|
| Real Algebraic Geometry                                                           | 31     |
| Representation Theory of Symmetric Groups                                         | 17     |
| Free Probability                                                                  | 14     |
| Schur-Weyl Duality                                                                | 14     |
| Matroid Theory                                                                    | 11     |
| Additive Combinatorics                                                            | 9      |
| Polymatroid Theory                                                                | 8      |
| Algebraic Geometry                                                                | 7      |
| Random Matrix Theory                                                              | 7      |
| Finite Geometry                                                                   | 6      |
| Noncommutative $L^p$ Geometry                                                     | 6      |
| Spectral Graph Theory                                                             | 6      |
| Plethysm Theory                                                                   | 6      |
| Ergodic Circuit Framework (communication complexity via dynamical systems)        | 4      |
| {“framework_name”: “Ergodic Circuit Framework”, “math_branch”: “COMM_COMPLEXITY”} | 4      |
| Fourier Analysis on Boolean Functions                                             | 4      |
| Algebraic Geometry of Secant Varieties                                            | 4      |
| FOURIER_ANALYSIS                                                                  | 4      |
| Fourier analysis of boolean functions                                             | 3      |

| Field                                                                                              | Cycles |
|----------------------------------------------------------------------------------------------------|--------|
| Algebraic Topology                                                                                 | 3      |
| matroid theory                                                                                     | 3      |
| Coarse Geometric Lifting (CGL) — Geometric Complexity Theory                                       | 3      |
| {“framework_name”: “Tropical Circuit Weight Analysis (TCWA)”, “math_branch”: “BOUNDED_ARITHMETIC”} | 3      |
| Tropical Circuit Weight Analysis (BOUNDED_ARITHMETIC)                                              | 3      |
| Schur-Weyl duality                                                                                 | 3      |
| Fourier Analysis of Boolean Functions                                                              | 3      |
| Persistent Homology                                                                                | 3      |
| Noncommutative Geometry                                                                            | 3      |
| Noncommutative Harmonic Analysis                                                                   | 3      |
| Noncommutative geometry                                                                            | 2      |

### Last 15 cycles (chronological)

| Time                 | Verdict      | Title                                                               |
|----------------------|--------------|---------------------------------------------------------------------|
| 2026-05-10 17:22 UTC | INCONCLUSIVE | Symmetric Group Fourier Coefficient Gap in Disjointness Communica   |
| 2026-05-10 17:37 UTC | INCONCLUSIVE | Harmonic Coefficient Sum Lower Bound for Resolution Proof Length    |
| 2026-05-10 18:03 UTC | INCONCLUSIVE | Noncommutative $L^p$ Norm Lower Bounds for Disjointness Communicati |
| 2026-05-10 19:08 UTC | INCONCLUSIVE | Cheeger Constant Exponentiates Resolution Length for Tseitin Form   |
| 2026-05-10 19:52 UTC | INCONCLUSIVE | Hilbert Function Inverse Proportional to 3-SAT Solution Count       |
| 2026-05-10 20:25 UTC | INCONCLUSIVE | Tropical Rank Gap in Read-Twice BP Transition Matrices              |
| 2026-05-10 20:31 UTC | INCONCLUSIVE | SOS Moment Matrix Rank and Max-CUT Integrality Gap                  |
| 2026-05-10 20:37 UTC | INCONCLUSIVE | Genus of Communication Curve Bounds Disjointness Complexity         |
| 2026-05-10 20:44 UTC | INCONCLUSIVE | Toric Variety Hilbert Function Bounded by SOS Degree for Max-CUT    |
| 2026-05-10 21:20 UTC | INCONCLUSIVE | Polymatroid Rank Lower Bound for Monotone k-CLIQUE                  |
| 2026-05-10 21:50 UTC | INCONCLUSIVE | Coxeter Polynomial Root Count Inverse Proportional to Resolution    |
| 2026-05-10 21:59 UTC | INCONCLUSIVE | Spectral Concentration of Moment Matrices in Max-CUT SOS Hierarchy  |

| Time                 | Verdict      | Title                                                                        |
|----------------------|--------------|------------------------------------------------------------------------------|
| 2026-05-10 22:09 UTC | INCONCLUSIVE | Noncommutative Fourier Coefficient Gap in Read-Twice BPs for IP <sub>2</sub> |
| 2026-05-10 22:42 UTC | INCONCLUSIVE | Additive Energy Inverse Proportional to Disjointness Discrepancy             |
| 2026-05-10 23:07 UTC | INCONCLUSIVE | Moment-Matrix Spectral Norm Inverse Proportional to Communication            |

## How to read the reports

- reports/supported\_findings.md — SUPPORTED conjectures with full test code. Paper-quality.
- reports/falsified\_findings.md — FALSIFIED with counterexamples.
- reports/daily\_YYYY-MM-DD.md — chronological log by day.
- pvsnp\_notebook.jsonl — raw JSONL, one entry per cycle (source of truth).

## PDF export

Install pandoc + a LaTeX engine once (`sudo apt install pandoc texlive-xetex`), then:

```
cd research/reports
for f in supported_findings.md falsified_findings.md notebook_summary.md; do
    pandoc "$f" -o "${f%.md}.pdf" --pdf-engine=xelatex -V geometry:margin=2cm
done
```